Akshata Madavi

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EDUCATION

San José State University — San Jose, CA
M.S. Software Engineering — May 2027

College of Engineering Pune (COEP) — Pune, India
B.S. Computer Engineering — May 2018

Relevant Coursework: Machine Learning, Deep Learning, Reinforcement Learning, AI Agents, Recommender Systems

SKILLS

Programming Languages: Python, SQL, JavaScript

Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost, NumPy, Pandas

Models & Methods: Transformers, Vision-Language-Action (VLA) models, behavior cloning/DAgger, LoRA, RAG, multi-agent systems, LLM evals

LLM & Agent Tooling: PydanticAI, LangChain, OpenAI-compatible APIs, Gemini API, Claude Code, E2B sandboxes

MLOps & Cloud: W&B Weave, MLflow, Logfire, Docker, GitHub Actions, GCP (Vertex AI, BigQuery), AWS, Azure

Data & Web: Spark, Kafka/KSQL, Airflow, Apache Iceberg, MySQL, SQLite, FastAPI, React, Node.js, Streamlit

EXPERIENCE

The Mortgage Office — San Mateo, CA
Applied AI Engineer Intern — Jul 2026 – Aug 2026

- Built an evaluation suite for production LLM agents with code-based and LLM-as-judge graders over ~1,000 test cases; caught an agent skipping required tool calls on ~50% of turns (fixed to 0%) and cut release review from weeks to days.
- Designed and shipped an agentic document-intelligence service (OCR, layout parsing, structured extraction) across 100+ document types, replacing a paid third-party API.

LTIMindtree — Pune, India
Senior Data Engineer — May 2023 – Mar 2025

- Redesigned a ~30 TB multi-source ETL service as parallel Spark jobs, cutting ingestion from 1 week to under 1 hour; with near real-time BigQuery pipelines (~10M records/run), it became the primary feature store for downstream ML models.

AI Adventures — Pune, India
Data Engineer — Aug 2021 – Apr 2023

- Led migration to an Apache Iceberg data lake, cutting storage costs ~50% and data latency ~95%; ran A/B tests and retention analysis on it that drove a 20% revenue increase.

STL (Sterlite Technologies) — Pune, India
Data Engineer — Jun 2018 – May 2021

- Consolidated ~1 TB of manufacturing data into BigQuery and trained an XGBoost model to pinpoint the highest-scrap process; the resulting root-cause fix cut scrap production by ~18%.

PROJECTS

Tumble – Autonomous ML Research Agent | *Python, FastAPI, React, E2B, W&B* — Sep 2026 – Present

- Built a coding-agent harness from scratch (ReAct loop, tool calling, sandboxed shell) and extended it into an agent that edits, runs, and scores its own training code via git keep/revert, raising a classifier from 96.5% to 98.3% accuracy.
- Productizing it as a multi-tenant app: E2B-sandboxed execution, session-isolated FastAPI backend, live-streaming React UI, and cost guardrails (per-IP rate limits, 10 experiments/session, global daily token budget).

VLA From Scratch – Vision-Language-Action Model | *PyTorch, DINOv2, MiniGrid/BabyAI* — Sep 2026 – Present

- Implemented an RT-2-style VLA in plain PyTorch (vision encoder, causal transformer, action-token head); raised closed-loop task success from 54% to 96% with DAgger, and to 87.5% with plain behavior cloning using a frozen DINOv2 encoder.
- Ported it to the BabyAI benchmark, where switching to an egocentric camera view lifted success from 18% to 53%; compared discrete vs. continuous action heads (54% vs. 68.5%).

Enterprise AI Governance – PII Detection | *PyTorch, Hugging Face, LoRA, MLflow, LangChain* — Oct 2025 – Feb 2026

- Fine-tuned Qwen 0.6B with LoRA, tuning rank/alpha/learning rate in MLflow, raising PII detection recall from 87% to 97% on a held-out set; served it through a Chrome extension for real-time detection in web forms.

LEADERSHIP & INVOLVEMENT

- **Open Source:** Published an eval-investigation skill for Claude Code/Codex that root-causes agent evaluation failures.
- **Hackathons:** Strange Loop (CoreWeave × W&B, 2026): self-auditing SWE-bench coding agent that catches reward hacking; VitalWatch (AI-Catalyst, 2025): real-time XGBoost anomaly detection, 98% precision, ~10 s alerts.
- **Peer Mentor:** Mentored 12 undergraduate students in Python, SQL, and data engineering.